Do festival features help electricity forecasting, and is the answer stable?

Typed versus pooled festival inputs for foundation and local models

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Research in progress

Why study this?

Festivals move

Diwali, Holi, Dussehra, Ganesh Chaturthi and Ugadi/Gudi Padwa change date every year and differ by state.

A simple design choice

Give each festival its own input (typed), or one shared "festival day" input (pooled)?

Precise target

Daily energy met/supplied in GWh: what the grid delivered, not unconstrained demand.

Data and evaluation rules

- Delhi, Karnataka, Maharashtra; event years 2024 and 2025; five festival families.
- 528 forecast cases = 176 state-dates × leads 1, 7, 14 (75 distinct dates).
- Effective event units: 10 festival-year clusters, not 528 independent tests.
- Past-only fitting, 730-day history, recorded labels only; NYC holidays as a separate-source check.

These outcomes were already inspected, so the results are exploratory, not a clean test.

Six models, eight matched input designs

Local models

Ridge regression

SARIMAX

LightGBM (+ leaf-2 variant)

Refit monthly, past data only

Foundation models

Chronos-2 and TimesFM-3

Pinned pretrained weights

No fine-tuning

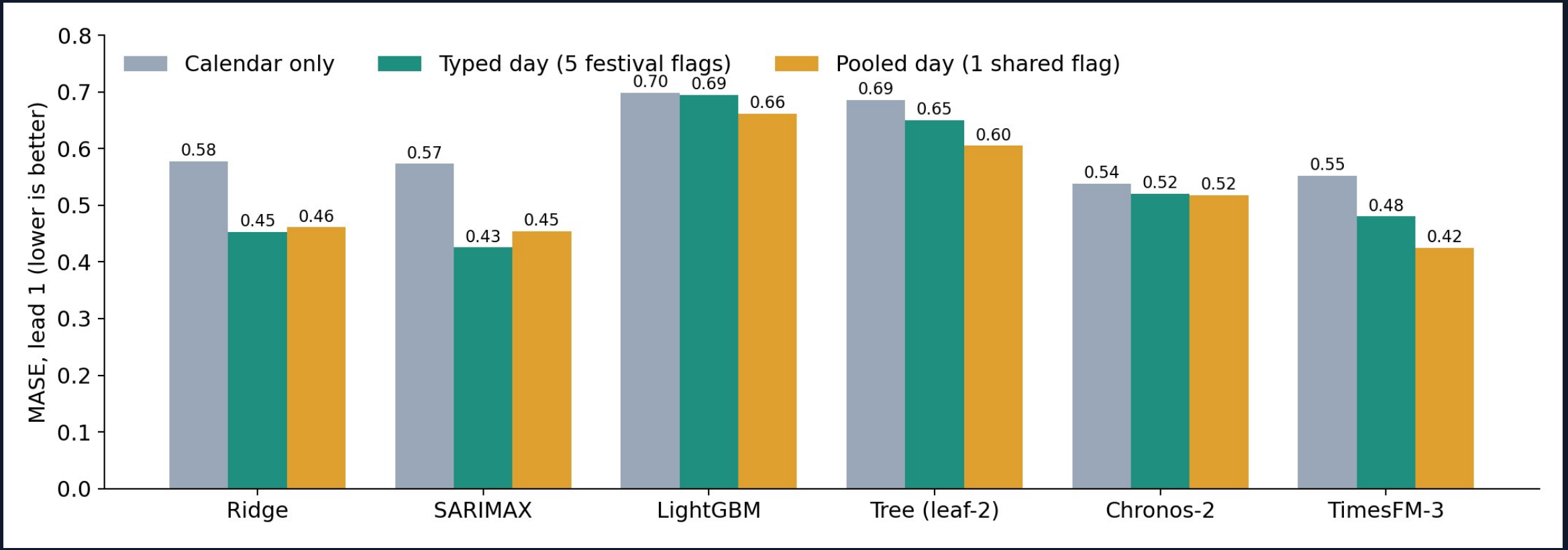
Input designs

Calendar only

Typed / pooled, day or ±3-day window

Zero channels and fake-date placebos as controls

Finding 1: festival flags help at lead 1



Lowest here: TimesFM-3 pooled (0.4246) and SARIMAX typed (0.4253).

Corrected Indian festival windows, 528 cases, lead 1. Source: v6 Table 1. No significance test; two event years.

Finding 2: the best encoding is not stable

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cells where 2024 and 2025 disagree on typed vs pooled (+1 exact tie)

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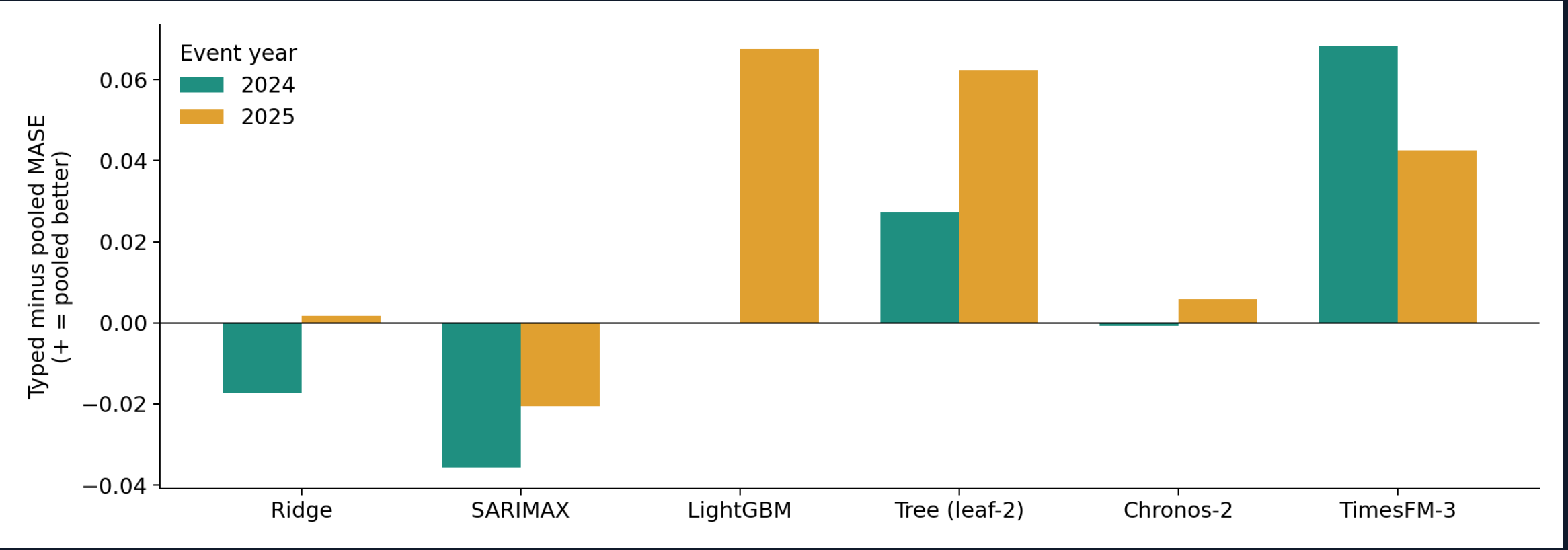
cells where dropping one festival flips the result

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cells where India and NYC disagree (+3 ties); all 6 TimesFM-3 cells reverse

Main lesson: report the ordering by year and source; do not pick one "winning" encoding.

Year by year: few orderings persist



SARIMAX favours typed in both years; TimesFM-3 favours pooled in 9/10 clusters.

Day flags, lead 1, typed minus pooled MASE by event year. Source: v6 Table A4.

Checks that changed the evidence

Calendar correction

Karnataka 2025: the notice names Vijayadashami on 1 Oct, not 2 Oct. The ledger was fixed from the document, not from errors.

Interval coverage

80% intervals cover less than 80% in 85 of 100 lead-1 festival clusters (Chronos-2 and TimesFM-3).

Honest counting

An exact-tie counting error was found and corrected; original outputs kept.

121 automated tests pass; this is software checking, not independent human review.

Closest prior work and what is new

Ziel (2018)

Compared one shared holiday dummy with separate dummies for German load: separate ones did not generally help.

Kreusel et al. (2026)

Calendar encodings (ordinal, one-hot, cyclic, target) in Chronos-2 and TimesFM-3, on hourly data.

This study

The same typed-vs-pooled choice through foundation-model inputs, with zero and placebo controls, across two sources.

Not a new architecture and not a causal holiday effect.

Confirmation on genuinely future data

1. Frozen forecasts

30–31 Oct 2026: forecasts saved publicly before outcomes. Score from 12 Nov (Diwali period included).

2. Weekly study

Registered: 52 weekly forecasts from 16 Nov. Starts only after rights and calendar checks.

3. Singapore replication

New country and festival calendar; protocol drafted, code ready, no data downloaded yet.

No future accuracy result exists yet.

What I can claim, and the limits

- Festival flags often reduce lead-1 error on these cases.
- Which encoding is best depends on model, year, festival set and source.
- Calendar provenance and native uncertainty need explicit checks.
- Limits: 2 event years, 3 related states, inspected data, rights reply pending, no external reviewer yet.

A careful, reproducible empirical study, still in progress.

Sources and questions

- Robbie Andrew, Indian Energy and Emissions Data (GRID-INDIA/POSOCO); reuse request pending.
- Hunt & Bloomfield, Zenodo 14983362 (CC BY 4.0); MTA NYC ridership; NESO demand.
- Ziel 2018, JMPSCE 6(2); Kreusel et al. 2026, arXiv 2610.13676; Chronos-2, arXiv 2510.15821.
- Code and results: github.com/Binay-verma/tsaf_project

Charts are generated from this project’s recorded results.